

Curriculum Vitae

Romulus Selepiuc

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Experience:

Senior Embedded Linux Engineer (contractor)

Period: 2021-07 <> present

Company and location: Essence Security, Israel (remote)

Responsibilities:

C/C++ Linux embedded software developer for security and safety systems.

Design, implement and test communication related features, using: TCP, WiFi, SMS, LTE, SMS, MQTT.

Streaming of data to designated devices.

Compression and encryption of transmitted data.

Senior Software Developer

Period: 2015-01 <> 2021-07

Company and location: Visteon Corporation, Sofia, Bulgaria

Project:

Daimler MFA2 (C++/Linux ARM32)

Responsibilities:

System performance and stability responsible

I have implemented solutions to reduce the boot time and to improve the overall system performance and stability. For example:

- kernel optimization (ex: removing features that are not used in our system)
- implementation of initrams for load of a minimal rootfs and early start-up of audio applications
- add security features (compiler and linker options, reduced application privileges, readonly partitions, and firewalls)
- automatic save of debug information in case of crash/watchdog timeout (coredumps, logs, journal, system info)
- system resources monitoring (print info or warnings logs related to system resource allocations)
- possibility to switch the target from release to debug mode with a CAN message (debug mode offers additional debug information, but the performance is affected)
- optimize the applications start-up sequences according to dependencies
- split the processes and interrupts across cpu cores to get better performance
- application crash analysis

Architecture, design and development of manufacturing module (C++/Linux)

- The manufacturing mode was designed as an independent mode, booting from a different partition and running different applications than normal mode.

Architecture, design and development of diagnostics module (C++/Linux)

Software Architect

Period: 2014-04 <> 2014-12

Company and location: CONTINENTAL AUTOMOTIVE ROMANIA SRL, Iasi

Responsibilities:

Requirements and change requests analysis

Software architecture and design

Technical support for the team.

Projects:

Peugeot - Navigation System

The project is running on embedded Linux platform ARM32.

I was responsible for Positioning area (communication with GPS chip, gyroscope sensor, speed sensor..).

Software Developer

Period: 2011-11 <> 2014-03

Company and location: CONTINENTAL AUTOMOTIVE ROMANIA SRL, Iasi, Romania

Responsibilities:

Software design

Software development(C++ on Linux and Windows Embedded, C# for test applications)

Projects

A phone application: The phone connects via Bluetooth to the car's system. On the car's display we can initiate or accept phone calls, access the phone book, and play the music stored on the phone.

I worked on this software for approximately a year and a half.

A web browser that is integrated with the car's infotainment system. The web browser implements a custom set of JavaScript APIs that provides information about the car (speed, consumption, GPS position ...).Also it connects to the internet. The applications was developed in C++ and runs on Windows Embedded Compact 7 platform. The browser is based on Qt / Webkit.

I worked on this software for approximately a year.

Consultant IT (Full time)

Period: 2008-03 <> 2011-07

Company and location: ELECTRICA SOLUZIONE SA, BUCHAREST

Responsibilities:

Software development (analysis, design, programming, implementation, documentation and maintenance) of electricity billing module for "E.ON Romania".

Technologies: C/C++, Power Builder, Oracle

IT Specialist /Network and database administrator (Full time)

Period: 2002-10 <> 2008-03

Company and location: MKB Romexterra Bank SA, Branch Bacau

Responsibilities:

Microsoft SQL Server administrator.

Bank software development (Delphi and C++ Builder/Windows)

Computer network administrator.

IT Specialist (Full time)

Period: 2001-02 <> 2002-06

Company and location: Raiffeisen Bank – Banca Agricola, Radauti Branch

Responsibilities:

Maintenance activities for banking applications.

Computer network administration

Software development

Analyst - programmer (Full time)

Period: 2000-09 <> 2001-02

Company and location: High School "Eudoxiu Hurmuzachi", Radauti

Responsibilities:

Software and web development

Computer network administration

Teaching activities

Education

University (graduate)

Technical University Gheorhe Asachi (1994 - 2000), Iasi, Romania

Major: Automatic, Computers

Skills

Programming: C, C++, C++11, Linux shell (bash), POSIX, Win32 API, MFC, Qt, Delphi, C++ Builder, Visual C++

Databases: Microsoft SQL Server, Oracle, SQL, PL/SQL, T-SQL

Tools: Rhapsody, Doors, RTC, Coverity, Git, Toad

Proficient in C++11, with a strong understanding of modern C++ features such as smart pointers, lambda expressions, and multithreading.

Proficient in Object-Oriented Design principles (encapsulation, inheritance, polymorphism).

Experience with concurrent programming and synchronization techniques (mutexes, semaphores, threads, critical sections).

Spoken foreign languages:

English - Advanced

Bulgarian - Advanced